

2(a)	(if in thermal contact) no <u>net transfer</u> of (thermal) energy (between them)	B1
2(b)(i)	$pV = nRT$	C1
	$T = (1.20 \times 10^5 \times 0.0260) / (0.740 \times 8.31)$ (= 507 K)	M1
	temperature = 507 – 273 = 234 °C	A1
2(b)(ii)	thermal equilibrium <u>so</u> temperatures (of X and Y) are equal	B1
	$pV = NkT$	C1
	$N = (2.90 \times 10^5 \times 0.0430) / (1.38 \times 10^{-23} \times 507)$ $= 1.78 \times 10^{24}$	A1
2(b)(iii)	(molecular) kinetic energy is proportional to temperature or kinetic energy (of molecules) is same in both cylinders - kinetic energy proportional to mass × mean-square speed or temperature proportional to mass × mean-square speed or r.m.s. speed proportional to $\sqrt{(\text{temperature} / \text{mass})}$ - mean-square speed inversely proportional to mass or r.m.s. speed inversely proportional to $\sqrt{(\text{mass})}$ <i>Any two bulleted points, 1 mark each</i>	B2
	r.m.s. speed (of molecules) in X is half r.m.s. speed (of molecules) in Y	B1